

```
import warnings
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import string
import re
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
warnings.filterwarnings("ignore")
```

```
data_fake = pd.read_csv("Fake.csv")
```

```
data_fake
```

	title	text	subject	date	
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017	  
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017	
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017	
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017	
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017	
...	...	...	...	...	
23476	McPain: John McCain Furious That Iran Treated ...	21st Century Wire says As 21WIRE reported earl...	Middle-east	January 16, 2016	
23477	JUSTICE? Yahoo Settles E-mail Privacy Class-ac...	21st Century Wire says It s a familiar theme. ...	Middle-east	January 16, 2016	
23478	Sunnistan: US and Allied 'Safe Zone' Plan to T...	Patrick Henningsen 21st Century WireRemember ...	Middle-east	January 15, 2016	

Next steps:

[Generate code with data_fake](#)[New interactive sheet](#)

```
data_true = pd.read_csv("True.csv")
```

data_true

	title	text	subject	date	
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017	  
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017	
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017	
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	politicsNews	December 30, 2017	
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...	politicsNews	December 29, 2017	
...	...	...	...	...	
21412	'Fully committed' NATO backs new U.S. approach...	BRUSSELS (Reuters) - NATO allies on Tuesday we...	worldnews	August 22, 2017	
21413	LexisNexis withdrew two	LONDON (Reuters) -	worldnews	August 22,	

Next steps:

[Generate code with data_true](#)[New interactive sheet](#)

```
data_true.tail()
```

	title	text	subject	date	
21412	'Fully committed' NATO backs new U.S. approach...	BRUSSELS (Reuters) - NATO allies on Tuesday we...	worldnews	August 22, 2017	
21413	LexisNexis withdrew two products from Chinese ...	LONDON (Reuters) - LexisNexis, a provider of l...	worldnews	August 22, 2017	
21414	Minsk cultural hub becomes haven from authorities	MINSK (Reuters) - In the shadow of disused Sov...	worldnews	August 22, 2017	
21415	Vatican upbeat on possibility of Pope Francis ...	MOSCOW (Reuters) - Vatican Secretary of State ...	worldnews	August 22, 2017	

```
data_fake.tail()
```

	title	text	subject	date	
23476	McPain: John McCain Furious That Iran Treated ...	21st Century Wire says As 21WIRE reported earl...	Middle-east	January 16, 2016	
23477	JUSTICE? Yahoo Settles E-mail Privacy Class-ac...	21st Century Wire says It s a familiar theme. ...	Middle-east	January 16, 2016	
23478	Sunnistan: US and Allied 'Safe Zone' Plan to T...	Patrick Henningsen 21st Century WireRemember ...	Middle-east	January 15, 2016	
23479	How to Blow \$700 Million: Al Jazeera America F...	21st Century Wire says Al Jazeera America will...	Middle-east	January 14, 2016	

```
data_true.head()
```

	title	text	subject	date
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	politicsNews	December 30, 2017

Next steps:

[Generate code with data_true](#)[New interactive sheet](#)

data_fake.head()

	title	text	subject	date
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn't wish all Americans ...	News	December 31, 2017
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day	News	December 25, 2017

Next steps:

[Generate code with data_fake](#)[New interactive sheet](#)

data_true.head()

	title	text	subject	date
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	politicsNews	December 30, 2017

Next steps:

[Generate code with data_true](#)[New interactive sheet](#)

data_fake.head()

	title	text	subject	date
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn't wish all Americans ...	News	December 31, 2017
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day	News	December 25, 2017

Next steps:

[Generate code with data_fake](#)[New interactive sheet](#)

```
data_fake["class"] = 0
data_true["class"] = 1
```

data_fake

	title	text	subject	date	class	
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn't wish all Americans ...	News	December 31, 2017	0	  
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017	0	
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017	0	
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017	0	
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017	0	
...	...	...	...	...	...	
23476	McPain: John McCain Furious That Iran Treated ...	21st Century Wire says As 21WIRE reported earl...	Middle-east	January 16, 2016	0	
23477	JUSTICE? Yahoo Settles E-mail Privacy Class-ac...	21st Century Wire says It's a familiar theme. ...	Middle-east	January 16, 2016	0	
23478	Sunnistan: US and ...	Patrick Henningsen	Middle-	January 15,	0	

Next steps:

[Generate code with data_fake](#)[New interactive sheet](#)

data_true

	title	text	subject	date	class	
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017	1	  
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017	1	
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017	1	
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	politicsNews	December 30, 2017	1	
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...	politicsNews	December 29, 2017	1	
...	...	...	...	...	...	
21412	'Fully committed' NATO backs new U.S. approach...	BRUSSELS (Reuters) - NATO allies on Tuesday we...	worldnews	August 22, 2017	1	

Next steps:

[Generate code with data_true](#)[New interactive sheet](#)

data_fake.shape, data_true.shape

((23481, 5), (21417, 5))

data_fake.sample()

	title	text	subject	date	class	
9617	OFF HER ROCKER! MAXINE WATERS: Trump Has 'Tend...	Rep. Maxine Waters (D., Calif.) Monday accused...	politics	Oct 24, 2017	0	

data_true.sample()

	title	text	subject	date	class	
18220	After British PM May's speech fiasco, her part...	LONDON (Reuters) - It was meant to be the spee...	worldnews	October 5, 2017	1	

```
data_fake_manual_testing = data_fake.tail(10)
for i in range(23480, 23470, -1):
    data_fake.drop([i], axis = 0, inplace = True)
data_true_manual_testing = data_true.tail(10)
for i in range(21416, 21406, -1):
    data_true.drop([i], axis = 0, inplace = True)
```

data_fake_manual_testing

	title	text	subject	date	class	
23471	Seven Iranians freed in the prisoner swap have...	21st Century Wire says This week, the historic...	Middle-east	January 20, 2016	0	
23472	#Hashtag Hell & The Fake Left	By Dady Chery and Gilbert MercierAll writers ...	Middle-east	January 19, 2016	0	
23473	Astroturfing: Journalist Reveals Brainwashing ...	Vic Bishop Waking TimesOur reality is carefull...	Middle-east	January 19, 2016	0	
23474	The New American Century: An Era of Fraud	Paul Craig RobertsIn the last years of the 20t...	Middle-east	January 19, 2016	0	
23475	Hillary Clinton: 'Israel First' (and no peace ...	Robert Fantina CounterpunchAlthough the United...	Middle-east	January 18, 2016	0	
23476	McPain: John McCain Furious That Iran Treated ...	21st Century Wire says As 21WIRE reported earl...	Middle-east	January 16, 2016	0	
23477	JUSTICE? Yahoo Settles E-mail Privacy Class-	21st Century Wire says It s a familiar theme. ...	Middle-east	January 16, 2016	0	

data_fake.shape, data_true.shape

((23471, 5), (21407, 5))

data_fake_manual_testing.head(10)

	title	text	subject	date	class
23471	Seven Iranians freed in the prisoner swap have...	21st Century Wire says This week, the historic...	Middle-east	January 20, 2016	0
23472	#Hashtag Hell & The Fake Left	By Dady Chery and Gilbert MercierAll writers ...	Middle-east	January 19, 2016	0
23473	Astroturfing: Journalist Reveals Brainwashing ...	Vic Bishop Waking TimesOur reality is carefull...	Middle-east	January 19, 2016	0
23474	The New American Century: An Era of Fraud	Paul Craig RobertsIn the last years of the 20t...	Middle-east	January 19, 2016	0
23475	Hillary Clinton: 'Israel First' (and no peace ...	Robert Fantina CounterpunchAlthough the United...	Middle-east	January 18, 2016	0
23476	McPain: John McCain Furious That Iran Treated ...	21st Century Wire says As 21WIRE reported earl...	Middle-east	January 16, 2016	0
23477	JUSTICE? Yahoo Settles E-mail Privacy Class-	21st Century Wire says It s a familiar theme. ...	Middle-east	January 16, 2016	0

```
data_true_manual_testing.head(10)
```

	title	text	subject	date	class
21407	Mata Pires, owner of embattled Brazil builder ...	SAO PAULO (Reuters) - Cesar Mata Pires, the ow...	worldnews	August 22, 2017	1
21408	U.S., North Korea clash at U.N. forum over nuc...	GENEVA (Reuters) - North Korea and the United ...	worldnews	August 22, 2017	1
21409	U.S., North Korea clash at U.N. arms forum on ...	GENEVA (Reuters) - North Korea and the United ...	worldnews	August 22, 2017	1
21410	Headless torso could belong to submarine journ...	COPENHAGEN (Reuters) - Danish police said on T...	worldnews	August 22, 2017	1
21411	North Korea shipments to Syria chemical arms a...	UNITED NATIONS (Reuters) - Two North Korean sh...	worldnews	August 21, 2017	1
21412	'Fully committed' NATO backs new U.S. approach...	BRUSSELS (Reuters) - NATO allies on Tuesday we...	worldnews	August 22, 2017	1
21413	LexisNexis withdrew two products from Chinese ...	LONDON (Reuters) - LexisNexis, a provider of l...	worldnews	August 22, 2017	1
21414	Minsk cultural hub becomes haven	MINSK (Reuters) - In the shadow of disused	worldnews	August	1

```
data_merge = pd.concat([data_fake, data_true], axis = 0)
```

```
data_merge.shape
```

```
(44878, 5)
```

```
data_fake
```

	title	text	subject	date	class
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn't wish all Americans ...	News	December 31, 2017	0
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017	0
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017	0
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017	0
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017	0
...	...	...	...	...	...
23466	Boston Brakes? How to Hack a New Car With Your...	21st Century Wire says For those who still ref...	Middle-east	January 22, 2016	0
23467	Oregon Governor Says Feds 'Must Act' Against P...	21st Century Wire says So far, after nearly 20...	Middle-east	January 21, 2016	0
23468	Ron Paul on Burns	21st Century Wire	Middle-	January 21,	0

Next steps:

[Generate code with data_fake](#)[New interactive sheet](#)

```
data_merge.head(10)
```

	title	text	subject	date	class	
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017	0	 
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017	0	
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017	0	
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017	0	
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017	0	
5	Racist Alabama Cops Brutalize Black Boy While...	The number of cases of cops brutalizing and ki...	News	December 25, 2017	0	
6	Fresh Off The Golf Course, Trump Lashes Out A...	Donald Trump spent a good portion of his day a...	News	December 23, 2017	0	
7	Trump Said Some INSANELY Racist Stuff	In the wake of yet another court decision	News	December	0	

Next steps: [Generate code with data_merge](#) [New interactive sheet](#)

```
data_merge.tail(10)
```

	title	text	subject	date	class
21397	Germany's Schulz says he would demand U.S. wit...	BERLIN (Reuters) - The leader of Germany's Soc...	worldnews	August 23, 2017	1
21398	Blunt instrument? What a list of banned articl...	SHANGHAI (Reuters) - An old review of an acade...	worldnews	August 23, 2017	1
21399	Saudi police release teenager detained for dan...	DUBAI (Reuters) - A 14-year-old boy who was de...	worldnews	August 22, 2017	1
21400	The People's Princess, Britons work to keep me...	LONDON (Reuters) - Abdul Daoud spilt most of t...	worldnews	August 23, 2017	1
21401	Argentina labor unions protest job losses, Mac...	BUENOS AIRES (Reuters) - Argentina's main labo...	worldnews	August 22, 2017	1
21402	Exclusive: Trump's Afghan decision may increas...	ON BOARD A U.S. MILITARY AIRCRAFT (Reuters) - ...	worldnews	August 22, 2017	1
21403	U.S. puts more pressure on Pakistan to help wi...	WASHINGTON (Reuters) - The United States sugge...	worldnews	August 21, 2017	1
21404	Exclusive: U.S. to withhold up to \$290	WASHINGTON (Reuters) - The United	worldnews	August	1

```
data_merge.columns
```

```
Index(['title', 'text', 'subject', 'date', 'class'], dtype='object')
```

```
data = data_merge.drop(['title', 'subject', 'date'], axis = 1)
```

```
data
```

	text	class	
0	Donald Trump just couldn't wish all Americans ...	0	
1	House Intelligence Committee Chairman Devin Nu...	0	
2	On Friday, it was revealed that former Milwauk...	0	
3	On Christmas day, Donald Trump announced that ...	0	
4	Pope Francis used his annual Christmas Day mes...	0	
...	...	...	
21402	ON BOARD A U.S. MILITARY AIRCRAFT (Reuters) - ...	1	
21403	WASHINGTON (Reuters) - The United States sugge...	1	
21404	WASHINGTON (Reuters) - The United States has d...	1	
21405	ISLAMABAD (Reuters) - Outlining a new strategy...	1	
21406	GENEVA (Reuters) - North Korea and the United ...	1	

44878 rows × 2 columns

Next steps:

[Generate code with data](#)

[New interactive sheet](#)

```
data.isnull().sum()
```

```

0
text    0
class   0
dtype: int64
```

```
data = data.sample(frac = 1)
```

```
data.head(10)
```

	text	class	
14466	COLOMBO (Reuters) - Sri Lankan police arrested...	1	
19712	BEIJING (Reuters) - China supports efforts by ...	1	
11801	NEW YORK (Reuters) - Lawyers for an executive ...	1	
6507	What could Dick Cheney, engineer of the wars t...	0	
2749	On Tuesday, Donald Trump announced the identit...	0	
19168	MEXICO CITY (Reuters) - Relief supplies and vo...	1	
12280	MOSCOW/WARSAW (Reuters) - Russian President Vl...	1	
6052	WASHINGTON (Reuters) - U.S. President Donald T...	1	
7229	BERLIN (Reuters) - European leaders and Presid...	1	
6192	There s been an ongoing battle between presump...	0	

Next steps:

[Generate code with data](#)[New interactive sheet](#)

data.tail(10)

	text	class	
18040	Gayle King was the first to ask Sanders if it ...	0	
9161	WASHINGTON (Reuters) - U.S. Secretary of State...	1	
13091	When Obama had the opportunity to speak out ag...	0	
3874	While the final vote tally won t be in for qui...	0	
16404	BAGHDAD/BEIRUT (Reuters) - Iraqi forces launch...	1	
17441	BEIRUT (Reuters) - The battle for the city of ...	1	
7446	Hacktivist group Anonymous has now commenced t...	0	
21203	SANTIAGO (Reuters) - Chilean President Michell...	1	
15532	George Soros is sticking his nose into voter l...	0	
1363	For a man who has spent his entire life fat-sh...	0	

```
def wordopt(text):
    text = text.lower()
    text = re.sub('[.*?\\]', '', text)
    text = re.sub("\\W", " ", text)
    text = re.sub('https?://\\S+|www\\.\\S+', '', text)
    text = re.sub('<.*?>+', '', text)
    text = re.sub('[%s]' % re.escape(string.punctuation), '', text)
```

```
text = re.sub('\w*\d\w*', '', text)
return text
```

```
data['text'] = data['text'].apply(wordopt)
```

```
data.head(10)
```

	text	class	
14466	colombo reuters sri lankan police arrested...	1	
19712	beijing reuters china supports efforts by ...	1	
11801	new york reuters lawyers for an executive ...	1	
6507	what could dick cheney engineer of the wars t...	0	
2749	on tuesday donald trump announced the identit...	0	
19168	mexico city reuters relief supplies and vo...	1	
12280	moscow warsaw reuters russian president vl...	1	
6052	washington reuters u s president donald t...	1	
7229	berlin reuters european leaders and presid...	1	
6192	there s been an ongoing battle between presump...	0	

Next steps:

[Generate code with data](#)
[New interactive sheet](#)

```
x = data['text']      # text data
y = data['class']     # labels
```

```
from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(
    x,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y)
```

```
from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer(
    stop_words='english',
    max_df=0.7
)
```



```
xv_train = vectorizer.fit_transform(x_train)
xv_test = vectorizer.transform(x_test)
```

```
from sklearn.linear_model import LogisticRegression

y_train = y_train.ravel()

LR = LogisticRegression(
    max_iter=2000,
    solver='lbfgs'
)

LR.fit(xv_train, y_train)
```

▼ LogisticRegression ⓘ ?
LogisticRegression(max_iter=2000)

```
y_pred = LR.predict(xv_test)
```

```
from sklearn.metrics import accuracy_score, classification_report

print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.9862967914438503
```

	precision	recall	f1-score	support
0	0.99	0.99	0.99	4694
1	0.99	0.99	0.99	4282
accuracy			0.99	8976
macro avg	0.99	0.99	0.99	8976
weighted avg	0.99	0.99	0.99	8976

```
print(xv_train.shape) # (same rows as y_train)
print(y_train.shape)
```

```
(35902, 97049)
(35902,)
```

```
from sklearn.tree import DecisionTreeClassifier

DT = DecisionTreeClassifier(
    criterion='gini',
    max_depth=20,
    random_state=42
)
```

```
DT.fit(xv_train, y_train)
```

▼ **DecisionTreeClassifier** ⓘ ?
 DecisionTreeClassifier(max_depth=20, random_state=42)

```
y_pred_dt = DT.predict(xv_test)
```

```
from sklearn.metrics import accuracy_score, classification_report

print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred_dt))
print(classification_report(y_test, y_pred_dt))
```

```
Decision Tree Accuracy: 0.9962121212121212
```

	precision	recall	f1-score	support
0	1.00	0.99	1.00	4694
1	0.99	1.00	1.00	4282
accuracy			1.00	8976
macro avg	1.00	1.00	1.00	8976
weighted avg	1.00	1.00	1.00	8976

```
from sklearn.linear_model import LogisticRegression

LR = LogisticRegression(max_iter=2000)
LR.fit(xv_train, y_train)
```

▼ **LogisticRegression** ⓘ ?
 LogisticRegression(max_iter=2000)